



Figure 1: Windows version 64-bit installation



Figure 2: WinPcap installation

The packet details pane gives all the encapsulation details of the information that flows down from the application layer to the physical layer. The bytes pane displays the contents of the packet selected as hexadecimal, ASCII formats.

Experiment 1

To understand the use of Wireshark, let us conduct a simple experiment. The aim is to check the connectivity from a desktop in a local area network, to its gateway. For the purposes of the experiment, let's assume that the gateway IPv4 address is 192.168.108.1, the source machine IP address is 192.168.108.83., and we use the command window of the source machine. The command ping is issued as follows:

```
C:\>ping 192.168.108.1 <Enter>.
```

Ping is a command line software utility used to check the connectivity from a source to a destination in a network. By default, ping sends four packets from the source to the destination. It uses the Internet Control Message Protocol (ICMP) to echo request packets and reply. The ICMP gives the results for the round-trip time and statistical information about the activity.

The ping command is used to echo the packets from the



Figure 3: Wireshark 2.2.5 (64-bit) installation completion

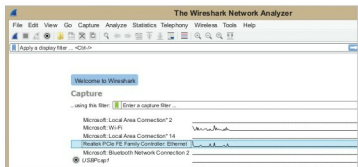


Figure 4: Interface selection, capture start window

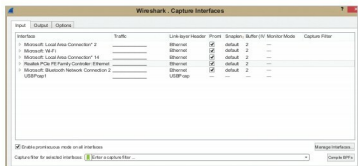


Figure 5: Capture options window

source to the destination, and the destination is 192.168.108.1. Four packets of size 32 bytes are sent serially to the destination. Wireshark sniffs these activities. To filter the ping results, the command syntax ICMP is applied in the filter available in Wireshark. It filters and displays the frames related to ICMP. Filters can be applied on protocols, fields, values, etc. To be more specific, the IPv4 address of the source machine is added to the filter.

```
Filter : icmp && ip.addr==192.168.108.78
```

The result is shown in Figure 8.

To validate the Wireshark capture, one of the frames is selected in the packet list pane (highlighted). The details with respect to the access layer, network layer, transport layer and application layer protocol data unit (PDU) encapsulation can be analysed.